

LiNbOCl₄ — crystalline oxyhalide benchmark

Testing transfer of the same workflow to mixed O²⁻/Cl⁻ chemistry

1 Literature

Tanaka et al. (2023)
10.4 mS cm⁻¹ at room T
Experimental E_a = 0.240 eV

2 Structure

Periodic oxyhalide model
224 atoms; 32 Li
MACE / SevenNet / M3GNet

3 MD protocol

600–1200 K; NVT
1 fs; 50 ps equilibration
500 ps production

4 Analysis

Li MSD and D(T)
ln(σNET) Arrhenius plot
300 K extrapolation

5 Conclusion

Experimental slope included
Model ordering differs
from Li₃YCl₆